The MYP Unit Planner

Teacher (s)	[Teacher names removed]	Subject group and discipline	Science		
Unit Title	Unit 3: Heat Transfer, Jet Streams, Weather and Climate	MYP Year	1	Unit Duration (hours)	50

Stage 1: Establishing the purpose of the unit

Key Concept: Select One:				Related Concept(s): 1, 2, or perhaps 3			Global Context: Select One		
Aesthetics	Change	Communication	Communities	Energy			Identities and relationships	Orientation in space and time	Personal and cultural expression
Connections	Creativity	Culture	Development	Interaction			Scientific and technical innovation	Globalization and sustainability	Fairness and development
Form	Global Interactions	Identity	Logic	movement			**Global Context Exploration** **Systems, models and methods**		
Perspective	Relationships	Systems	Time, Place, and Space						

Statement of inquiry The **statement of inquiry** focuses the purpose/goal of the unit. Use the global context of your choice (title of GC or the descriptor), the key concept, and the related concepts to form the statement of inquiry. The statement of inquiry should be written prior to the inquiry questions.

Conceptual Understanding: (Key concept, related concept, connect with verb)	SOI:
Systems interact with one another through the movement of energy.	Every day weather conditions can impact the routines and decisions of people.

Inquiry questions The **inquiry questions** is how we encourage looking at this topic through multiple perspectives in order to lead students through the metacognitive process from academic knowledge to thoughtful action. This helps to develop positive attitudes and a sense of personal and social responsibility. It is helpful to have anywhere from three to five inquiry questions that will guide students through the unit. Use your related concepts to help guide you in this process. Inquiry questions provide structured inquiry into factual, conceptual and procedural knowledge that leads to higher-order thinking.

Factual	Conceptual	Debatable
Knowledge/fact-based Content-driven Skills-related Supported by evidence Can be used to explore terminology in the statement of inquiry Frequently topical Encourage recall and comprehension	Enable exploration of big ideas that connect facts and topics Highlight opportunities to compare and contrast Explore contradictions Lead to deeper disciplinary and interdisciplinary understanding Promote transfer to familiar or less familiar situations, issues, ideas and contexts Encourage analysis and application	Enable the use of facts and concepts to debate a position Promote discussion Explore significant ideas and issues from multiple perspectives Can be contested Have tension May be deliberately provocative Encourage synthesis and evaluation
• What type of breeze happens during the day time? • What type of breeze happens during the night time? • Does cold air have high or low pressure? • Does warm air have high or low pressure? • What are the spheres of the Earth's Systems?	• What is convection and how is it related to weather? • What is convection and how is it related to breezes? • How does the sun start weather patterns? • Which spheres does the water cycle need in order to continue?	• Which of Earth's spheres is the most important sphere? • Is knowing the weather or climate more important? • Can Earth still function if one system is in critical condition?

Commented [E1]: This should be vague and incorporate the key, related concept, and global context. Consider: "Systems created by energy, interaction, and movement can facilitate scientific and technical innovation"

MYP Criteria & Objectives	Outline the summative assessment task	
Criterion A: Knowing and understanding i. outline scientific knowledge; ii. identify and comment upon the creator's choices iii. interpret information to make scientifically supported judgments.	Unit 3 IBIS Paint Project Design an ibis paint image of a location that has: • During the day or the night • Has least 3 of the 5 earth spheres present (must have geosphere and hydrosphere). • over the hydrosphere and geosphere, must have convection arrows that demonstrates the heat flow of convection over the water and the land (land breeze or sea breeze) Written portion: 1. explain your location and its general yearly climate. 2. Describe today's weather in this location. a. Temperature, air pressure, precipitation levels. 3. Explain why there is sea breeze during the day and land breeze at night. 4. Explain the elements of each Earth sphere you included in your design. 5. BONUS: What was the biggest challenge for you during this project? How did you overcome it? Name one thing you can do to celebrate your success.	Relationship between summative assessment task(s) and statement of inquiry: The image they will create will allow them to solidify their understanding of the Earth's spheres and their various interactions, as well as how the weather is impacted due to the time of day.

Approaches to Learning: Pick 1-2 areas to focus on. Use the pattern for aligning ATL skill indicator (PiP pg. 98) with objective. You may also include these skills in the learning process categories.

ATL Categories	Thinking	Social	Communication	Self-Management	Research
ATL Clusters	Critical thinking Creative thinking Transfer	Collaboration	Communication	Organization Affective Reflection	Information Literacy Media Literacy
In order for a student to (objective strand)			Give and receive meaningful feedback.	In order for students to decide what they will wear today, students must	

Commented [E2]: Consider adding more Criteria and objectives.

Commented [E4]: This should be written in first person to describe how your Statement of Inquiry will be addressed with your summative.

Commented [E3]: Is this able to be adjusted to fit the GRASPS model?

_____ , **students must (skill indicator, p.98)** _____.**Or a narrative to explain how/when you will teach this.**			Students will give and receive meaningful feedback using the video Daniel's butterfly and then practicing voicing what students want for feedback and other students will practice giving feedback for growth.	understand how to read predicted weather patterns for the day and the week.	

Commented [E5]: This section seems Incomplete. Can you highlight which ATL cluster will be addressed.

Action: Teaching and learning through inquiry

Learning Process *Learning experiences and teaching strategies. Connect to content and ATL skills.*				
Progression of Learning	Content: Objective/Skill/Standards	Learning Experience/ Activities	Teaching Strategies/ Differentiation	Formative Assessment/Feedback
Day one	TN Standard Skill Objective Question	Kids are going to do	What am I going to do Who am I going to help: Struggle bus, and the fast bus	Check for understanding, how will I give students feedback?
Week 1	Frontloading Questions	Review videos, benchmarks and vocabulary words		Teacher will go over answers and provide correct definition
Week 1	Articles and Questions (heat transfer, Earth's spheres)	Deepen understanding of initial concepts		Teacher will go over answers
Week 2 and 3	STEMscopes			Teacher will go over answers
Week 4	IPad ibisPaint + Forms questions	Kids will fill in the blanks and answer questions from ibisPaint drawing and Microsoft Forms Quiz	Tag students in chat so they know what they're responsible for in the interactive whole-class interaction	Teacher will go over answers
Week 5	Kahoot and Quizizz			Review
Week 5	Mastery Connect			

Commented [E6]: This section is incomplete. Please fill in.

Resources Tools and resources, tools and equipment, information, learning environment, local environment, invitation to expert from the community
STEMscopes.com, STEMscopes bok, Brainpop, Youtube, Discovery.com, classroom charts, Microsoft Forms, CANVAS discussions

Reflection: Considering the planning, process and impact of the inquiry.

Prior to teaching the unit	During teaching	After teaching the unit
Why do we think that the unit or the selection of topics will be interesting? What do students already know, and what can they do? What have students encountered in this discipline before? What does experience tell us about what to expect in this unit? What attributes of the learner profile does this unit offer students opportunities to develop? What potential interdisciplinary connections can we identify? What do we know about students' preferences and patterns of interactions? Are there any possible opportunities for meaningful service learning? What in the unit might be inspiring for community or personal projects? Could we develop authentic opportunities for service learning? How can we use students' multilingualism as a resource for learning?	What difficulties did we encounter while completing the unit or the summative assessment task(s)? What resources are proving useful, and what other resources do we need? What student inquiries are emerging? What can we adjust or change? What skills need more practice? What is the level of student engagement? How can we scaffold learning for students who need more guidance? What is happening in the world right now with which we could connect teaching and learning in this unit? How well are the learning experiences aligned with the unit's objectives? What opportunities are we hearing to help students explore the interpretative nature of knowledge including personal biases that might be retained, revised or rejected? (DP theory of knowledge skills development)	What were the learning outcomes of this unit? How well did the summative assessment task serve to distinguish levels of achievement? Was the task sufficiently complex to allow students to reach the highest levels? What evidence of learning can we identify? What artifacts of learning should we document? Which teaching strategies were effective? Why? What was surprising? What student-initiated action did we notice? What will we do differently next time? How will we build on our experience next unit? How effectively did we differentiate learning in this unit? What can students carry forward from this unit to the next year/level of study? Which subject groups could we work with next time? What did we learn from standardizing the assessment?

Commented [E7]: This section should include first-person reflection that answers SOME of the prompt questions presented.